33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10006

F13-SR-005
415V MILL CHILLERS & REAGENTS SWITCHROOM

F13-TR-020
TRANSFORMER ROOM

F13-TR-020
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF

SWR EARTH BAR
Tx EARTH BAR
NOTE 3
4000/ 1A
2.5VA
5P20

F13-SW-020
MILL CHILLERS & REAGENTS 415V MOTOR CONTROL CENTRE

MCC EARTH BAR
NEUTRAL BAR
1/700A
NOTE 1
1000/ 1A
2.5VA
5P20

4000/ 5A
30VA
C10.5
TEST
PM
20A
63A
2A
20A
240VAC SPRING CHARGE SUPPLY
ACB
2A
AFD
50G
4000A
50 50N 50G
51 51N 51G
24VDC CONTROL POWER SUPPLY
NOTE 7
4000 /5A
50VA
C11.0
SD
NOTE 6
100A
100A
INC

415V 3NE~ 50Hz 4000A 65kA for 1sec

01F-01 1000A 2A 32A 400A
1F-01 1000A 2A 32A 400A
01R-01 1000A 2A 32A 400A
1R-01 1000A 2A 32A 400A
02F-11 630A 2A 32A 400A EOL
2F-11 630A 2A 32A 400A EOL
04R-13 400A 2A 32A 170A

1
FEEDER TO F13-SW-120
2000-F13-SLD-EL-10711
2000-F13-SCM-EL-10716
LOAD MOVED FROM SHEET 4

2R-17 100A 2A 32A 400A EOL
02F-05 160A 2A 32A 400A EOL
2F-05 160A 2A 32A 400A EOL
04R-01 20A 2A 32A 70A EOL
1
AHF

E-SHUT
F13SR005-LCS
NOTE 12

CONTINUED ON SHT 2
2000-F13-SLD-EL-10702

1 CP 454kW F13-CU-402 CHILLER 1 2000-F13-SCM-EL-10708
CP 454kW F13-CU-403 CHILLER 2 2000-F13-SCM-EL-10709
CP 454kW F13-CU-404 CHILLER 3 2000-F13-SCM-EL-10710
CP 454kW F13-CU-405 CHILLER 4 2000-F13-SCM-EL-10711
M 3~ 185kW F13-PW-413 COOLING WATER PUMP 1 2000-F13-SCM-EL-10712
M 3~ 185kW F13-PW-414 COOLING WATER PUMP 2 2000-F13-SCM-EL-10713

ACTIVE HARMONIC FILTER
F13-HF-020
100A
1 2000-F13-SCM-EL-10707

M 3~ 45kW F13-PW-415 CHILLER RECIRCULATING WATER PUMP 1 2000-F13-SCM-EL-10714
M 3~ 45kW F13-PW-416 CHILLER RECIRCULATING WATER PUMP 2 2000-F13-SCM-EL-10715
1
M 3~ 2.2kW F72-AG-465 FLOCCULANT MIXING TANK AGITATOR 2000-F72-SCM-EL-10701

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 24.06.2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F13-SW-020 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F13-BLK-IC-10004.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

DRG No.	REFERENCES
2000-F13-SCM-EL-10703	F13-SW-020 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM
2000-F13-SCM-EL-10702	F13-SW-020 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM
2000-F13-SCM-EL-10701	F13-SW-020 415V MCC INCOMER SCHEMATIC DIAGRAM
2000-F13-DRG-EL-10121	F13-SR-005 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F13-BLK-EL-10341	F13-SR-005 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM
2000-F13-SLD-EL-10702	F13-SW-020 415V MCC SINGLE LINE DIAGRAM SHEET 2
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 5
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	29MAY25	FAT REDLINES INCORPORATED	[redacted]				
0	25MAR24	APPROVED FOR CONSTRUCTION					

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: MILLING & CLASSIFICATION
F13-SW-020 415V MCC
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F13-SLD-EL-10701	1

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